

PS1 - IO

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1 - Data Import and Descriptive Statistics

```
# Load necessary libraries
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(readr)
library(psych)
```

Attaching package: 'psych'

The following objects are masked from 'package:ggplot2':

%+%, alpha

```
library(tidyr)
library(knitr)
```

```
# Import the dataset
df <- read_delim("~/SchoolWork/Y2S1/I0/PS1/IRI.csv")
```

Rows: 63966 Columns: 18

-- Column specification -----

Delimiter: "\t"

chr (5): market_name, state, brand, parent, segment

dbl (13): year, month, store_id, week_id, flavored, fortified, fiber, sugars...

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
glimpse(df)
```

Rows: 63,966

Columns: 18

```
$ year      <dbl> 2008, 2008, 2008, 2008, 2008, 2008, 2008, 2008, 2008, 2008~
$ month     <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4~
$ store_id  <dbl> 369, 108, 374, 221, 294, 117, 378, 310, 237, 321, 120, 275~
$ week_id   <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5~
$ market_name <chr> "BOSTON", "BOSTON", "BOSTON", "BOSTON", "BOSTON", "BOSTON"~
$ state     <chr> "Massachusetts", "Massachusetts", "Massachusetts", "Massac~
$ brand     <chr> "KELLOGGS RAISIN BRAN CRUNCH", "GENERAL MILLS GOLDEN GRAHA~
$ parent    <chr> "KELLOGGS", "GENERAL MILLS", "KELLOGGS", "GENERAL MILLS", ~
$ flavored  <dbl> 1, 1, 1, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1~
$ fortified <dbl> 0, 1, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1~
$ fiber     <dbl> 3.759398, 0.000000, 0.000000, 0.000000, 0.000000, 2.000000~
$ sugars    <dbl> 9.022556, 9.000000, 3.000000, 13.000000, 3.000000, 8.00000~
$ price     <dbl> 0.1680204, 0.2733062, 0.1723915, 0.2528735, 0.3389252, 0.2~
$ quantity  <dbl> 600.6000, 96.0000, 1889.2100, 165.1040, 60.0000, 224.0000,~
$ puma      <dbl> 2700, 2700, 2700, 2700, 2700, 2700, 2700, 2700, 2700, 2700~
$ sugar_price <dbl> 211.9174, 211.9174, 211.9174, 211.9174, 211.9174, 211.9174~
$ M         <dbl> 117679.10, 55841.92, 204192.90, 29956.03, 35946.88, 96805.~
$ segment   <chr> "Family", "Family", "Child", "Child", "Child", "Adult", "A~
```

```
describe(df$brand)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis
X1*	1	63966	19.96	11.28	20	19.95	14.83	1	39	38	0	-1.21
	se											
X1*	0.04											

```
describe(df$parent)
```

```
vars      n mean  sd median trimmed  mad min max range skew kurtosis se
X1*      1 63966 2.12 1.09      2    2.03 1.48   1  4     3 0.58   -0.97  0
```

2 - Descriptive Statistics by Brand

```
cereal_vars <- c("price", "brand_num", "sold",
                "parent_num", "sugars", "fiber", "flavored", "flavored",
                "fortified")

df_market <- df %>%
  mutate(
    brand_num = n_distinct(brand),
    parent_num = n_distinct(parent),
    sold = (price * quantity)
  ) %>%
  group_by(week_id, store_id) %>%
  summarise(
    across(all_of(cereal_vars),
      list(
        mean = ~ mean(.x, na.rm = TRUE),
        sd = ~ sd(.x, na.rm = TRUE),
        median = ~ median(.x, na.rm = TRUE),
        max = ~ max(.x, na.rm = TRUE),
        min = ~ min(.x, na.rm = TRUE),
        n = ~ sum(!is.na(.x))),
      .names = "{.col}_{.fn}") %>%
    pivot_longer(everything(),
      names_to = c("variable", "stat"),
      names_sep = "_",
      values_to = "value")
```

`summarise()` has grouped output by 'week_id'. You can override using the
`.groups` argument.

Warning: Expected 2 pieces. Additional pieces discarded in 12 rows [9, 10, 11, 12, 13, 14, 21, 22, 23, 24, 25, 26].